Electric jug

The challenge

The challenge in this activity is to try to increase the temperature of 200 mL of tap water as much as possible in five minutes using the equipment provided.

The equipment available to your group

A 12 volt power supply, leads with alligator clips, different types of wire, a 200 mL beaker, a thermometer and a light globe. .

1. Work in groups and spend five minutes designing your experiment.
   You must include the light globe in your circuit to prevent the power pack from over heating.

2 Carry out your first trial experiment and record your results in a table.

3. List ways that you could improve your design.

4. Carry out your second trial experiment and record your results in a table.

5. Draw a circuit diagram to represent the circuit.

Electric kettles and jugs in the home work on the same principle as the electric jug you have designed.

6. List two reasons why thin wire is wrapped in coils around the ceramic support in an electric jug.

7 Why do you think the jug's body is ceramic or plastic and NOT metallic ?

8 Why is nichrome wire rather than iron wire more likely to be used ?